Report: Final Project Phase 1

Group N

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1. Introduction

This report covers how we anonymized the survey dataset (Step 2), the analyses we ran on the anonymized dataset (Step 3) and our reflection on the trade-off between disclosure risk and utility. The dataset has 200 respondents. After anonymization, the exact date of birth is no longer available, and age is only given in four bands: 18-29, 30-44, 45-59 and 60+. The analyses use age, sex, education, the party the respondent voted for (Red or Green) and the voting channel (evote = 0 for paper voting at a polling station, evote = 1 for online voting). Two respondents did not vote Red or Green, so the party analyses use 198 people. We used a 5% significance level for all tests. Three types of analysis were done: (A) a comparison of the survey with the certified results, (B) the relationship between party choice and demographics, and (C) the relationship between voting channel and demographics. For B and C we used chi-square tests of independence for each variable separately, followed by a logistic regression with all three demographic variables together.

2. Anonymization and Disclosure Risk

2.1 Disclosure scenario

We assume an adversary who has our anonymized dataset, the public population register, the published election results and this report, and nothing else. The adversary's goal is to learn how a named person voted. They would do so by linking a released record to a person in the register.

We classified the survey attributes by how they can be used for linking (Table 1):

Table 1. Classification of the survey attributes

Attribute	Also in the register?	Role
Name	Yes	Direct identifier
Sex, dob, zip, marital_status, citizenship	Yes	Quasi-identifier
Evote	Yes (as last_voted)	Quasi-identifier
Education	No	Not linkable with the available data
Party	No	Sensitive attribute

The voting channel is a quasi-identifier because the register records how each person voted (paper, electronic or abstained). Since survey respondents all voted, the adversary can restrict the register to 1113 voters. Education does not appear in any public file, so it cannot be used for linking under our scenario.

The register covers the whole population. Population frequencies of each attribute combination can therefore be counted exactly rather than estimated.

2.2 Baseline risk of the raw data

Without anonymization, 199 of the 200 respondents (99.5%) can be re-identified. We measured this by removing only the name and linking each record to the 1113 voters in the register on sex, date of birth, ZIP code and voting channel.

For each respondent we counted k, the number of voters in the register who share the same quasi-identifier values. A respondent with $k = 1$ is unique and is re-identified with certainty. We also computed the expected number of re-identifications as the sum of $1/k$ over all respondents. This assumes the adversary picks one candidate at random from each matching group.

In the raw data, the expected number of re-identifications is 199.5. The exact date of birth is the main driver: on its own it is unique for most people in the register.

2.3 Anonymization methods

We combined variable removal, global recoding and local suppression, using a target of $k \geq 3$ for every respondent. Global recoding was applied first because it treats all records the same way. Local suppression was then used only for the few records that remained risky.

Every recoding was applied identically to the survey and to the register before measuring risk. This mirrors what an adversary can do, since they read the same recoding rules in this report.

1. **Removal**. The name was removed. The exact date of birth was removed after deriving age from it.
2. **Age**. Age was computed at the date of data collection (1 July 2025) and grouped into four bands: 18–29, 30–44, 45–59 and 60+. Bands are wide enough to break uniqueness but still allow age to be analyzed in analyses B and C.
3. **Citizenship**. Recoded to Denmark or Other. Almost all respondents are Danish, and most other citizenships occurred only once or twice in the survey, so each country on its own was close to an identifier.

4. **Marital status**. Divorced and Widowed were merged into Divorced/Widowed, because Widowed was a small category. Never married and Married/separated were kept.
5. **Local suppression**. For every record with $k < 3$ after recoding, we blanked citizenship and measured risk again. Records still at $k < 3$ then had marital status blanked. A blank cell is treated as matching any value, so suppression always increases k.

Sex, ZIP code, voting channel, education and party were released unchanged. We did not perturb party, so the survey's party distribution is preserved exactly for analysis A.

2.4 Disclosure risk after anonymization

After anonymization, no respondent is unique and every respondent matches at least 3 voters in the register. Table 2 shows how the risk fell at each stage. The expected number of re-identifications fell from 199.5 to 24.8, an 88% reduction. Risk was measured on sex, age band, ZIP code, voting channel, marital status and citizenship. This is a stricter test than the baseline, which only used sex, date of birth, ZIP code and voting channel, so the 88% reduction is if anything understated.

Table 2. Disclosure risk at each stage of anonymization

Stage	Unique respondents (k=1)	Respondents with k<3	Expected re-identifications
Raw data (name removed)	199	200	199.5
After global recoding	11	19	37.3
+ citizenship suppressed	4	8	29.9
+ marital status suppressed	0	0	24.8

The loss of information from suppression is small: 27 cells were blanked (19 citizenship, 8 marital status). That is 1.7% of the 1600 values in the released file. No records were removed, so all 200 respondents remain available for analysis.

We deliberately do not report which attribute combinations were unique or which records were suppressed. Publishing them would point an adversary directly at the riskiest records.

2.5 Limitations

Our protection holds under the stated scenario but has three known weaknesses.

- Choice of k. With $k \geq 3$, an adversary who guesses at random within a matching group is right at most one time in three. A higher k (for example 5) would lower this further,

but would require more suppression or coarser age bands, and therefore more utility loss.

- Attribute disclosure (l-diversity). k-anonymity protects identity, not the vote. If all released records in one group report the same party, the adversary learns that party without knowing which record belongs to the target. Party is unevenly split in the survey, so such homogeneous groups do occur. We did not perturb party, because doing so would directly distort analysis A.
- Leaked list of respondents. Our risk measure assumes the adversary does not know who took part in the survey. If that list leaked, the adversary would only need to distinguish between 200 respondents instead of 1113 voters. Many records would then be unique again. Protecting against this would require much coarser data or random perturbation of party (for example PRAM), at a considerable cost to analyses A–C.

3. Results of the Step 3 Analysis

3.1 Survey compared with the certified election results

First we checked whether the survey matches the official results. For each voting channel we ran a chi-square goodness-of-fit test, comparing the number of Red and Green voters in the survey with the numbers expected from the certified results. In the certified results, 35.8% of paper votes and 35.3% of online votes went to Red.

Table 3. Survey party choice compared with certified results

Channel	N	Observed (Red/Green)	Expected (Red/Green)	Chi-square	p-value
Paper	147	46 / 101	52.7 / 94.3	1.32	0.251
Online	51	21 / 30	18.0 / 33.0	0.78	0.378

Expected counts are the certified shares multiplied by the survey sample size.

Neither test in Table 3 is significant, so the survey does not differ from the certified results in either channel. The survey has a slightly lower Red share for paper voting (31.3%) and a slightly higher one for online voting (41.2%), but these differences are small enough to be explained by chance in a sample of this size. This analysis does not use age, so anonymization had no effect on it.

3.2 Party choice and demographics

Next, we looked at whether party choice is related to age, sex and education, using chi-square tests of independence. Because age is now in bands, we could not use a t-test on mean age as with exact ages.

Table 4. Party choice by age band

Age band	Green	Red	% Red
18-29	32	1	3.0%
30-44	34	10	22.7%
45-59	36	20	35.7%
60+	29	36	55.4%

Chi-square = 29.98, df = 3, p < 0.001.

There is a strong link between age and party. The share of Red voters goes up in every older band, from 3.0% among 18-29 year olds to 55.4% among people aged 60 and over (Table 4).

Women voted Red more often than men: 50 of 115 women (43.5%) compared with 17 of 83 men (20.5%). The difference is significant (chi-square = 10.38, df = 1, p = 0.001).

Education is also related to party (chi-square = 36.60, df = 8, p < 0.001). The Red share is highest among people with a vocational bachelor's degree (66.7%) and PhD (5 of 5), and lowest among those with upper secondary (12.5%) or primary education (17.1%) (Table 5). However, several education groups have very few people (for example only 5 with a PhD and 3 with 'Not stated'), so many expected counts are below 5 and this p-value should be treated as approximate.

Table 5. Party choice by education level

Education	Green	Red	% Red
Primary education	34	7	17.1%
Upper secondary education	14	2	12.5%
Vocational education and training (VET)	47	18	27.7%
Short cycle higher education	8	2	20.0%
Vocational bachelor's	11	22	66.7%
Bachelor's programmes	4	2	33.3%
Master's programmes	11	8	42.1%
PhD programmes	0	5	100%
Not stated	2	1	33.3%

Chi-square = 36.60, df = 8, p < 0.001.

To see the effects together, we fitted a logistic regression with Red as the outcome and age band, sex and education as predictors. Education groups with fewer than 5 respondents were merged into one group. The main results are in Table 6.

Table 6. Logistic regression for voting Red (odds ratios)

Predictor	Odds ratio	95% CI	p-value
Age 30-44 (vs 18-29)	8.30	0.84 - 81.57	0.070
Age 45-59 (vs 18-29)	20.42	2.12 - 196.40	0.009
Age 60+ (vs 18-29)	46.27	5.00 - 427.82	< 0.001
Male (vs female)	0.20	0.08 - 0.47	< 0.001

N = 198, pseudo R-squared = 0.33. Education terms are not shown; none were significant (all p > 0.24).

After controlling for the other variables, older age groups still have much higher odds of voting Red, and men have about 80% lower odds than women. Education is no longer significant once age and sex are in the model, which suggests that the education pattern in the simple test is partly explained by age and sex differences between the education groups. Two technical problems should be mentioned. The model gave a convergence warning, and the PhD odds ratio is huge with an infinite upper limit, because all 5 PhD holders voted Red (perfect separation). That odds ratio cannot be interpreted. The confidence intervals for the age bands are also very wide, partly because the reference group (18-29) only has one Red voter.

3.3 Voting channel and demographics

We did the same type of analysis with the voting channel as the outcome. Overall 53 of the 200 respondents (26.5%) voted online. There is no significant relationship between channel and age (chi-square = 1.27, df = 3, p = 0.736), sex (chi-square = 0.00, p = 1.000) or education (chi-square = 9.91, df = 8, p = 0.271). The share voting online was almost the same for women (26.7%) and men (26.2%) (Figure 1), and it varied between 23% and 32% across the age bands.

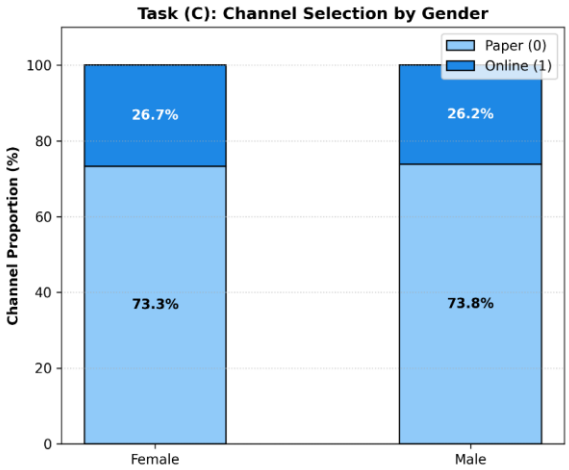


Figure 1. Channel selection by gender.

The logistic regression for online voting agrees with this. The model is not significant overall (likelihood ratio p = 0.70, pseudo R-squared = 0.03) and none of the predictors are significant. The age band odds ratios are all below 1 (between 0.57 and 0.82 compared with 18-29), which would mean older people are a bit less likely to vote online, but the confidence intervals include 1 so this could be chance.

4. Reflection on disclosure risk and utility

To judge the utility loss, we compared the Step 3 results with the same analyses on the original data from Step 1. The conclusions are the same in all three analyses. The survey still matches the certified results (identical, since party and channel were not changed). Age, sex and education are still linked to party, and men still have clearly lower odds of voting Red (odds ratio 0.20 now, 0.21 before). For the voting channel, nothing is significant in either version. Table 7 summarizes the main comparisons.

Table 7. Key results before and after anonymization

Result	Original data (Step 1)	Anonymized data (Step 3)
Survey vs certified (paper, online)	p = 0.251, p = 0.378	p = 0.251, p = 0.378
Age vs party	Mean age 58.7 vs 45.8, p < 0.001	Chi-square p < 0.001
Sex vs party	p = 0.001	p = 0.001
Odds ratio, male voting Red	0.21	0.20
Age vs voting channel	Welch p = 0.154	Chi-square p = 0.736
Sex, education vs channel	p = 1.000, p = 0.271	p = 1.000, p = 0.271

There are some differences, mostly because of the age bands:

With exact ages we could say that Red voters were on average about 13 years older (58.7 vs 45.8 years) and that each extra year raised the odds of voting Red by about 4%. Now we can only say that older bands are more likely to vote Red, and we cannot give means or use a t-test.

The odds ratios for the age bands have very wide confidence intervals (for example 5.0 to 427.8 for 60+), so the size of the age effect is quite uncertain, although the direction is clear.

For voting channel, exact age came closer to significance (Welch t-test $p = 0.15$, logistic regression $p = 0.077$) than the bands ($p = 0.74$ for the chi-square test). Neither is significant, so the conclusion is the same, but the bands may hide a small tendency for younger people to vote online.

The pseudo R-squared for the party model went up slightly (0.28 to 0.33), but we do not see this as an improvement because the model now has three age parameters instead of one.

Overall, we think the trade-off is good enough. The anonymization lowered the risk of respondents being re-identified (the expected number of re-identifications fell by 88%, from 199.5 to 24.8), while all the main findings stayed the same. The loss of information was mostly limited to how precisely age can be described, and only 27 cells (1.7% of all values) had to be suppressed. The utility loss is small partly because only age was generalized among the variables used in these analyses. If someone wanted to study marital status or citizenship, which were also generalized or suppressed, the loss could be bigger. Likewise, anyone needing exact age effects, for example a specific age threshold, would find four bands too coarse. A middle option would be narrower bands, such as ten-year groups, which keep more detail but would also keep more risk, so we would only use them if the risk measures allowed it.

There are also limits to these conclusions that have nothing to do with anonymization. The sample is small (200 people), so the tests can only find fairly large effects, and 'no significant relationship' for voting channel does not prove that there is none. Several education groups are small, which made the chi-square tests approximate and caused the separation problem for PhD holders. Finally, this is observational survey data, so the results show associations and not causes.